Department of Computer Science
Gujarat University

MCA112 Mathematical Foundation

Assignment – III

Unit 3 Graph Theory & Unit 4 Fundamentals of Vectors

Question 1. Define the following terms with example:

1) Graph 2) loop 3) Parallel edges
4) Adjacent vertices 5) Adjacent edges 6) Simple graph
7) Isolated vertex 8) k-regular graph 9) Degree of Vertex
10) Subgraph 11) Complete graph 12) Path
13) Eccentricity 14) Spanning subgraph
15) Radius of graph 16) Vertex deleted subgraph 17) Cycle
18) Diameter of Graph 19) Edge deleted subgraph
20) Tree 21) Self-complementary graph 22)Complement of graph
23) Connected graph 24) Adjacency Matrix 25) Acyclic graph
26) Isomorphic Graph

Question 2. State and prove First theorem of Graph theory

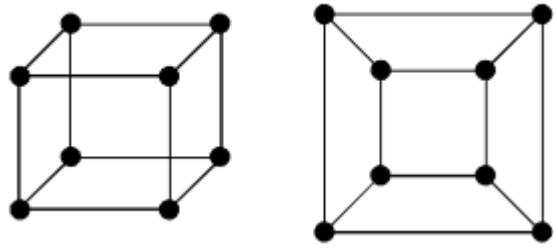
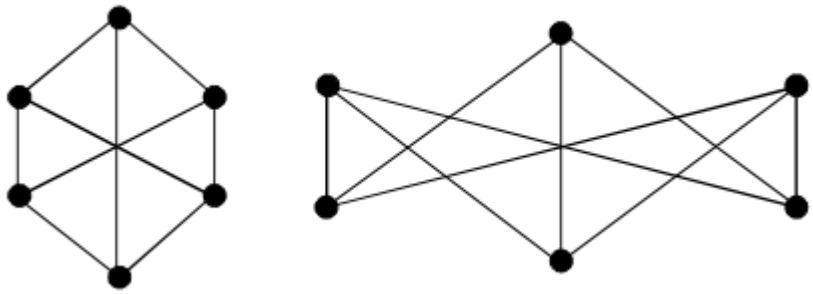
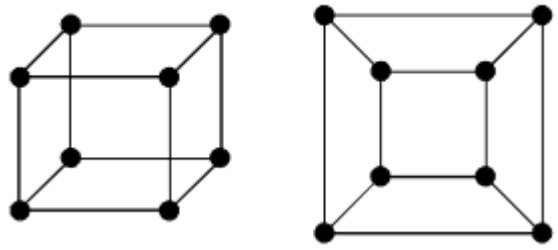
or

State and prove Handshaking theorem for graph.

Question 3. Construction of the graph based on conditions:

1) Construct two graph having same degree sequence
2) Construct two simple graph having same degree sequence
3) Construct two regular graph having same degree sequence
4) Construct the two complete graph having different degree sequence
5) Construct two regular graph having different degree sequence
6) Construct at least two graph having 6 vertices, 2 of degree 4 and 4 of degree 2.

Question 4. Show that the following graphs are isomorphic:

1)	
2)	
3)	

Question 5. Define following terms:

1) Vector 2) Scalar 3) Norm 4) Unit vector

5) Angle between two vectors 6) Orthogonal set

7) Orthonormal set 8) Projection v along w

9) Orthogonal Complement 10) Dot product in $\mathbb{R}^n$

Question 6. Normalize each of the following vectors:

1) $2i + 3j$ 2) $4i - 2\mathrm{j}$

3) $2i - j + 2k$ 4) $-i + 2j - 4k$

Question 7. Find dot product and angle between given vectors u and v:

1) $u = 2i + 3j, \quad v = 5i + 2j$

2) $u = i + j - k, \quad v = 3i - j + k$

3) $u = 3i + j + 2k, \quad v = 2i + 3j + 4k$

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Question 8. Find distance between each other and projection of u on v for following vectors:

1) $u = i - 3j + 4k, \quad v = 3i + 4j + 7k$
2) $u = i - 2j + 3k, \quad v = 2i + 4j + 5k$

Question 9. Express the following vectors as sums of scalar multiples of $u = (1, 1, 1), v = (1, 1, 0), w = (2, 0, 3)$:

1) $6i + j + 3j$
2) $3i - 2j - 3k$
3) $2i + j + k$

Question 10. Use the Gram-schmidt process to orthonormalise the set of linearly independent vectors:

1) $\{(1, 2, 1), (-1, 1, 0), (5, -1, 2)\}$
2) $\{(1, 0, 1, 1), (-1, 0, -1, 1), (0, -1, 1, 1)\}$
3) $\{(1, -1, 2, 0), (1, 1, 2, 0), (3, 0, 0, 1), (2, 1, -1, 0)\}$

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